Field Reflection Journal I Template

Mathematical Standards Observed
All standards cited from Michigan Department of Education (n.d.).

- **Kindergarten** — Michigan Department of Education (n.d.) identifies on page 13 from the section Counting and Cardinality the standard of knowing number names and the count sequence, including that students should count to 100 by ones and by tens, count forward beginning from a given number within the known sequence, and write numbers from 0 to 20 and represent a number of objects with a written numeral 0–20.
- **1st Grade** — Michigan Department of Education (n.d.) identifies on page 17 from the section operations and algebraic thinking the standard of representing and solving problems involving addition and subtraction, including that students use addition and subtraction within 20 to solve word problems involving situations of adding to, taking from, putting together, taking apart, and comparing, with unknowns in all positions, and solve word problems that call for addition of three whole numbers whose sum is less than or equal to 20.
- **4th Grade** — Michigan Department of Education (n.d.) identifies on page 31 from the section operations and algebraic thinking the standard of using the four operations with whole numbers to solve problems, including that students interpret a multiplication equation as a comparison, multiply or divide to solve word problems involving multiplicative comparison, and solve multistep word problems posed with whole numbers using all four operations.
- **5th Grade** — Michigan Department of Education (n.d.) identifies on page 37 from the section operations and algebraic thinking the standard of writing and interpreting numerical expressions, including that students use parentheses, brackets, or braces in numerical expressions, and evaluate expressions with these symbols, and write simple expressions that record calculations with numbers, and interpret numerical expressions without evaluating them.

Details of observations as of the end of Module Three: (If you have multiple field sites, include the information for each.)	**School:** **Mcbain Elementary School** **Teachers:** **Mrs. Vandervelde (Kindergarten) · Mrs. Avery (1st Grade & 4th Grade) · Mr. Dallas Milligan (5th Grade)** **Grade level:** **Kindergarten, 1st Grade, 4th Grade, 5th Grade** **Date(s):** **April 21, 2026 · April 22, 2026** **Duration of time for each observation:** **April 22 — 2 hours (Kindergarten) · April 22 — 3 hours (1st Grade) · April 21 & 22 — 1 hour (4th Grade) · April 21 — 4 hours (5th Grade) · Total: 10 hours**
Building Brave Mathematicians	**16 points**
1. Describe instances where students were encouraged to take risks, make conjectures, or explore alternative strategies in solving mathematical problems.	During my observation in Mrs. Vandervelde's kindergarten classroom on April 22, students were working on counting sequences and representing numbers with written numerals. When a student said she was not sure how to write the number 14, Mrs. Vandervelde did not give her the answer. Instead, she encouraged the student to try it first on her whiteboard. When the student wrote 41 instead of 14, Mrs. Vandervelde responded warmly, saying, Look at you — you used both of those digits! Let's think

	together about which one comes first. This is a clear example of encouraging risk-taking. In Chapter Three in the Taking Risks section, Dance and Kaplan (2018) explain that students need to feel confident enough to try new strategies and share their thinking even when it contradicts the thinking of others, and that this kind of confidence can only grow inside a classroom culture where wrong answers are treated as starting points rather than endpoints. A second example occurred in Mr. Dallas Milligan's 5th grade classroom on April 21, where students were working on writing and interpreting numerical expressions with parentheses. Mr. Milligan posed the problem 2 × (8 + 7) and asked students to try solving it a different way than they had before. One student volunteered to come to the board and attempt a strategy using repeated addition. Mr. Milligan praised the attempt and used it as a class discussion point. In Chapter Three in the Building Brave Mathematicians section, Dance and Kaplan (2018) establish that bravery in mathematics means returning to a problem after a setback, attempting an unfamiliar strategy, and believing that the struggle itself is the learning.
2. How did the teacher create a safe and supportive learning environment for students to engage in productive struggle and persist through challenging tasks?	In Mrs. Vandervelde's kindergarten classroom, the physical environment itself communicated safety. Student work was displayed on the walls — including work that showed crossed-out attempts and corrections — communicating that imperfect thinking was welcome. When one student grew frustrated during a counting activity and put his head down, Mrs. Vandervelde crouched beside him and said quietly, that feeling you have right now means your brain is working really hard. That's a good thing. In Chapter Three in the Mathematical Confidence: Why It Matters section, Dance and Kaplan (2018) explain that children with low self-confidence may get angry and give up when they feel they are not smart enough to figure a problem out, and that it is the

	classroom's responsibility to replace that fear with brave, supported persistence. A second example came from Mr. Milligan's 5th grade room on April 21, where he had established a classroom norm of silent thinking time for the first two minutes of any new problem. Students were expected to attempt something on their paper before any discussion began, ensuring every student engaged with the task before hearing a peer's approach. In Chapter Three in the Perseverance section, Dance and Kaplan (2018) note that it is the teacher's job to help students gain perseverance by building consistent classroom structures and expectations that treat productive struggle as a normal and valued part of learning.
3. Reflect on how the teacher's language and feedback promoted a sense of mathematical confidence and a willingness to engage in mathematical discourse.	Language was one of the most powerful tools I observed across all four classrooms. In Mrs. Vandervelde's kindergarten room, she consistently praised effort rather than correctness. When students completed their number writing practice, she would say things like, I love how hard you worked on that or I can see you tried three different times — that is being a mathematician. In Chapter Three in the Language Use section, Dance and Kaplan (2018) identify that praise must target effort and process rather than intelligence, because students praised for being smart begin to protect that identity by avoiding challenge. A second and especially memorable language moment occurred in Mrs. Avery's 1st grade classroom on April 22. A student said, I can't add those three numbers together, and without missing a beat, Mrs. Avery replied, you can't — yet. She then guided the student to use linking cubes to model the problem. In Chapter Three in the Student Language section, Dance and Kaplan (2018) explain that when teachers respond to the phrase I can't by simply adding the word yet, students begin to see success not as a two-path road where you either get it or you do not, but as a

	bumpy, windy, hilly trail that can be pushed through.
4. Provide examples of how the teacher celebrated students' effort, perseverance, and growth mindset rather than solely focusing on correct answers.	In Mrs. Avery's 4th grade classroom, students were working on multistep word problems using multiplication and division. One student had set up her equation incorrectly but had drawn a detailed diagram showing her thinking. Mrs. Avery held up the paper — with the student's permission — and said, look at this diagram. This student showed me exactly what she was thinking. That is what mathematicians do. She then used the diagram to guide the class in identifying where the thinking needed to shift, without ever framing the student's work as simply wrong. In Chapter Three in the Valuing Mistakes section, Dance and Kaplan (2018) argue that in a math classroom, mistakes should be celebrated as great mistakes and learning opportunities, not looked at as simply wrong answers. A second example came from Mr. Milligan's 5th grade classroom on April 21. After a student spent nearly five minutes working through a problem involving brackets and parentheses and arrived at an incorrect answer, Mr. Milligan said, five solid minutes of not giving up. That is what this class is about. He then asked the student to walk the class through his process, reinforcing that the journey mattered more than the destination. In Chapter Three in the Perseverance section, Dance and Kaplan (2018) further explain that children with mathematical confidence are able to persevere through challenging problems, trying again and again until they figure them out, and that it is the teacher's job to cultivate that perseverance over time.
Great Minds Think Differently	**16 points**
1. Describe situations where students were encouraged to approach problems from different perspectives or employ multiple strategies.	In Mr. Milligan's 5th grade classroom on April 21, after students solved a numerical expression problem, Mr. Milligan asked, did anyone solve this a different way than the person next to you?

	Three students shared three entirely different approaches — one used the standard order of operations, one drew a visual model, and one wrote out the steps in words before computing. He placed all three approaches side by side on the board. In Chapter Four in the Multiple Strategies section, Dance and Kaplan (2018) explain that when students are exposed to many strategies and representations, they learn to pick the strategy that not only makes the most sense but is also the most efficient given a particular problem, and that comparing strategies strengthens mathematical understanding. A second example occurred in Mrs. Avery's 4th grade classroom, where students were asked to solve the same word problem using both a drawing and an equation. This requirement pushed all learners to think flexibly rather than relying on a single mode of representation. In Chapter Four in the opening section, Dance and Kaplan (2018) establish that there are different strategies to solve a problem and that students should be flexible in using a variety of them, and that while finding the answer is the end goal, it is more important to be able to explain how you got there.
2. How did the teacher accommodate and value diverse thinking processes and representations among students?	In Mrs. Vandervelde's kindergarten classroom on April 22, students were not required to represent their numbers in the same way. Some used written numerals, some drew dots, and some used both. Mrs. Vandervelde acknowledged each representation equally and asked students to explain what they had done. In Chapter Four in the Justification section, Dance and Kaplan (2018) describe justification as the act of explaining not just what was done but why, and note that when students focus on the process, they develop understanding because they are at the center of explaining, providing evidence, generalizing, and articulating connections. A second example came from Mrs. Avery's 1st grade room on April 22, where students solving addition word problems

	used tally marks, number lines, and drawn pictures. Rather than redirecting students to one method, Mrs. Avery built the lesson's discussion around the variety, asking students to identify similarities between their approaches. In Chapter Four in the Student Shares section, Dance and Kaplan (2018) explain that student shares allow learners to see similarities and differences between each other's strategies and make meaningful connections that deepen understanding of the underlying mathematical concept.
3. Provide examples of how the teacher facilitated discussions that allowed students to share their unique problem-solving approaches and learn from one another.	Mr. Milligan regularly used structured sharing time in his 5th grade classroom. After independent work time, he called on students to explain their full process, and when one student finished, he asked the class, can someone restate what she just said in your own words? This technique pushed students to listen actively and internalize their classmates' thinking. In Chapter Four in the Student Shares section, Dance and Kaplan (2018) advocate for student shares where classmates explain each other's strategies in their own words, because this reinforces active listening and deepens understanding of multiple approaches. A second example came from Mrs. Avery's 4th grade room, where she used a strategy gallery after students solved a multistep problem. Students posted their work on the board and the class walked around silently reading each other's papers before any discussion began. In Chapter Four in the Reading Thinking section, Dance and Kaplan (2018) describe this exact technique — having students examine a classmate's paper silently and try to reconstruct the strategy used before any explanation is given — noting that it reveals whether student work is truly clear and teaches students to evaluate the efficiency of their own strategy choices.
4. Reflect on how the teacher's instructional practices celebrated and leveraged the diversity	In Mr. Milligan's room on April 21, after a student used an unconventional strategy to evaluate a

of students' mathematical thinking and experiences.	bracketed expression, Mr. Milligan named the strategy after the student for the rest of the lesson, writing Marcus's method on the board. This public naming validated the student's unique contribution and gave the whole class a shared vocabulary for the approach. In Chapter Four in the Public Records section, Dance and Kaplan (2018) encourage the creation of living anchor-chart posters that document students' strategies by name, giving students ownership and a daily reference for the classroom's growing mathematical thinking. A second example occurred in Mrs. Avery's 1st grade class on April 22. When two students arrived at the same answer through entirely different approaches — one through drawing and one through a number line — Mrs. Avery said, Isn't it amazing that two different paths led to the same place? That's what math is. She then asked those two students to explain their strategies side by side. In Chapter Four in the Multiple Strategies section, Dance and Kaplan (2018) note that teachers must create a classroom culture that empowers students to bravely compare their strategies with those of their classmates, because students who find connections between different approaches understand both more deeply than students who used only one.
Thinking Through Questioning	**16 points**
1. Observe and record examples of questions the teacher asked to promote deeper thinking, reasoning, and understanding among students.	Mr. Milligan's 5th grade classroom on April 21 provided the richest examples of deliberate questioning observed across all ten hours. During the lesson on writing and interpreting numerical expressions, he consistently asked questions such as what do we know about this problem and why did you use that strategy rather than evaluating student responses with praise or correction. In Chapter Five in the Types of Questions section, Dance and Kaplan (2018) identify three categories of teacher questions — those that clarify and probe for justification, those that

	guide, challenge, and extend thinking, and those that assess understanding — and explain that a skilled teacher moves fluidly between all three within a single conversation. A second example came from Mrs. Avery's 4th grade room, where she asked a student who had arrived at an incorrect answer, can you use a second strategy to prove it rather than pointing out the error directly. This question naturally led the student to discover the inconsistency in her own work. In Chapter Five in the Questions That Guide, Challenge, and Extend section, Dance and Kaplan (2018) explain that these types of questions are used to push student thinking, often while creating cognitive dissonance, and that they allow students to deepen their own reasoning as they justify their thinking and navigate through misconceptions.
2. How did the teacher's questioning strategies encourage students to explain their thought processes, justify their reasoning, and make connections between mathematical ideas?	In Mrs. Vandervelde's kindergarten classroom on April 22, she frequently asked students can you tell me more about that after a student gave an answer. This simple open-ended question consistently prompted students to add detail, use manipulatives to demonstrate their thinking, or connect their answer to a real-world example. In Chapter Five in the Questions That Clarify and Probe Thinking section, Dance and Kaplan (2018) explain that clarification questions help students understand a task, make sense of a problem, and explain their thinking more clearly, and that probing questions encourage students to think about a problem at a deeper level. A second example occurred in Mrs. Avery's 1st grade room on April 22. After students solved an addition word problem, she asked, how is your strategy similar to what your partner did? This question required students to look across strategies and identify mathematical connections rather than treating each solution as isolated. In Chapter Five in the Types of Questions section, Dance and Kaplan (2018) explain that strategic, deliberate questions can push student thinking to deeper

	levels and open up opportunities for quick formative assessment, making questioning not a supplement to mathematics instruction but its backbone.
3. Describe instances where the teacher used questioning to uncover students' misconceptions or gaps in understanding and provide appropriate scaffolding or support.	In Mrs. Avery's 4th grade class, during the lesson on interpreting multiplication equations as comparisons, several students were confusing multiplicative and additive comparisons. Rather than reteaching directly, Mrs. Avery asked the class, does that strategy always work and then offered a simpler example where the additive approach would break down. Through questioning alone, students were able to recognize that their reasoning was incomplete. In Chapter Five in the Questions That Guide, Challenge, and Extend section, Dance and Kaplan (2018) explain that these questions help students who are stuck find an entry point into a problem and allow students to navigate through misconceptions on their own terms rather than through direct correction. A second example occurred in Mr. Milligan's 5th grade room on April 21. When a student evaluated 2 × (8 + 7) by ignoring the parentheses and computing left to right, Mr. Milligan did not correct him. Instead, he asked, what does that number mean in the problem? He then asked what were we trying to figure out first? Through questioning alone, the student re-examined the expression and self-corrected. In Chapter Five in the Questions That Assess Understanding section, Dance and Kaplan (2018) explain that questions used as formative assessment allow teachers to make real-time instructional decisions about what to revisit, who to challenge, and where misconceptions are hiding.
4. Reflect on how the teacher's questioning techniques fostered a classroom culture of inquiry, curiosity, and intellectual discourse around mathematical concepts and ideas.	What struck me most across all four classrooms was how the teachers' questioning habits had transferred to the students themselves. In Mr. Milligan's 5th grade room on April 21, I observed students asking each other why did you use that

	strategy and can you explain that in your own words during partner work — language that clearly originated from the teacher's own questioning patterns. In Chapter Five in the Questioning: Why It Matters section, Dance and Kaplan (2018) establish that as teachers question students, they are also modeling the exact questioning techniques students will eventually use with each other as they analyze and compare their classmates' mathematical reasoning. A second example came from Mrs. Vandervelde's kindergarten class on April 22. During a counting activity, one student spontaneously asked a classmate, how did you know to start there — an unprompted, inquiry-driven question from a five-year-old. In Chapter Five in the Questioning: Why It Matters section, Dance and Kaplan (2018) further emphasize that becoming an artful questioner requires patience and practice, and that teachers must truly listen to students and be responsive to where they are in their understanding, because students who observe that responsiveness daily begin to replicate it in their own peer conversations.
Takeaways for Future Practice	**12 points**
1. Reflect on how the observed practices align with the mathematical teaching practices outlined in the Michigan Mathematical Practice Standards.	The practices I observed across all four classrooms aligned closely with the Michigan Mathematical Practice Standards in several important ways. The consistent emphasis on explaining thinking and justifying reasoning connects directly to the practice of constructing viable arguments and critiquing the reasoning of others. In every classroom, students were expected not simply to arrive at a correct answer but to demonstrate how and why they reached it. In Chapter Four in the Justification section, Dance and Kaplan (2018) argue that justification makes thinking visible and gives teachers the evidence they need to assess understanding and catch misconceptions before they become habits. Additionally, the multi-strategy approach used across all grade levels aligns with the practice of making sense of problems and persevering in solving them. Michigan Department of Education

	(n.d.) outlines on page 31 that fourth graders are expected to solve multistep word problems and assess the reasonableness of answers using mental computation and estimation strategies including rounding — skills that require exactly the kind of flexible, reflective thinking that all four teachers in these classrooms were actively cultivating.
2. Discuss areas where you felt the teacher could further enhance their practices to support building brave mathematicians, valuing diverse thinking processes, and promoting deeper mathematical reasoning through questioning.	While the teaching I observed was strong overall, one area across several classrooms where there was room for growth was wait time. On multiple occasions, teachers asked a deep, open-ended question and then called on a student within just a few seconds. This practice can inadvertently privilege students who process quickly and discourage those who need more time to formulate their thinking. In Chapter Five in the Teacher Talk Moves, Modeling, and Praise section, Dance and Kaplan (2018) emphasize that teachers must be flexible with their questioning because even with the best-planned lesson, a teacher can never quite know which direction a conversation may flow, and that becoming an artful questioner demands that teachers truly listen to students and be responsive to where they are in their understanding. Extending wait time to at least three to five seconds before calling on a student would give all learners a more equitable chance to engage. A second area for growth involves celebrating mistakes more explicitly and publicly. While teachers in these classrooms responded kindly to incorrect answers, I did not observe any teacher explicitly naming a student's mistake as a gift to the whole class in a celebratory, public way. In Chapter Three in the Valuing Mistakes section, Dance and Kaplan (2018) argue that mistakes should be celebrated as great mistakes and learning opportunities rather than simply handled quietly and moved past.
3. Identify aspects of the observed practices that you found particularly effective or inspiring and	The practice I found most inspiring and that I am most committed to bringing into my own future classroom is the deliberate use of questioning

would like to incorporate into your future teaching.	over statement-making. In Mr. Milligan's 5th grade class especially, I was struck by how rarely he told students anything directly. Almost everything was posed as a question, and the result was a room full of students who were visibly thinking rather than passively receiving. In Chapter Five in the Questions vs. Statements section, Dance and Kaplan (2018) note that when a teacher states the answer or explains the strategy, the cognitive load shifts from the student to the teacher, and that a question invites a student into thinking while a statement closes the door on it. I want to practice preparing specific, purposeful questions before each lesson rather than relying on in-the-moment improvisation. A second practice I intend to carry into my own teaching is the naming of student strategies publicly on anchor charts. In Chapter Four in the Public Records section, Dance and Kaplan (2018) encourage the creation of living anchor-chart posters that document students' strategies by name, giving students ownership and a daily reference for the classroom's growing mathematical thinking. Both of these practices reflect the kind of teacher I want to become — one who builds mathematical confidence, honors diverse thinking, and uses questions to open doors rather than close them.

References

Dance, R., & Kaplan, T. (2018). *This is not a math book: Building mathematical dispositions in young learners.* Heinemann.

Michigan Department of Education. (n.d.). *Michigan k-8 mathematics standards.* https://www.michigan.gov/-/media/Project/Websites/mde/Literacy/Content-Standards/Math_Standards.pdf?rev=1e793e2b1e314e4fa1abc754251b5dc9